ECE 332 Spring 2026

Exam 1 Solutions

(100 pts total)

May 7, 2026

Allowed: Calculator approved for the FE exam and a 2-sided page of notes.
Not allowed: Any other devices or materials.

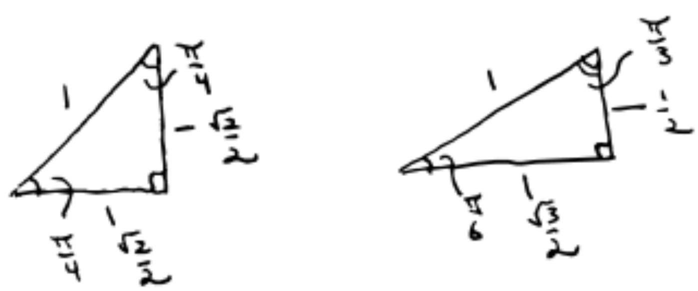
Use the approximations

$$\epsilon_0 \approx \frac{1}{36\pi}\times 10^{-9}\ \mathrm{F/m}, \quad \mu_0 = 4\pi\times 10^{-7}\ \mathrm{H/m}, \quad \eta_0 \approx 120\pi\ \Omega, \quad c \approx 3\times 10^{8}\ \mathrm{m/s}$$

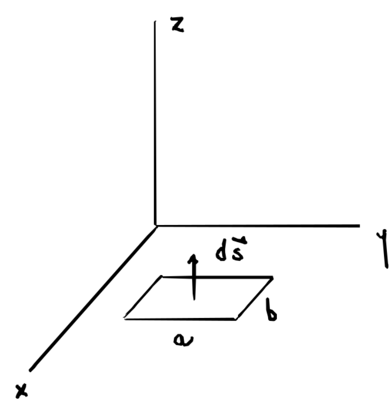
in your calculations.

Useful Facts:

"Special" right triangles:



1. (20 points total) Suppose a rectangular conducting loop of N turns with sides of length a parallel to the y-axis and sides of length b parallel to the x-axis is located in the x-y plane with normal $d\mathbf{s} = \hat{\mathbf{z}}ds$, as shown, in a magnetic field



$$\mathbf{B}(t) = B_0(\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + 3\hat{\mathbf{z}})\cos(\omega t).$$

(a) Find a formula for the magnetic flux through one turn of the loop. Simplify as much as possible. (6 points)

The flux through one turn in the loop is

$$\begin{aligned}\Phi &= \int_S \mathbf{B}\cdot d\mathbf{s} = \int_S (B_0(\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + 3\hat{\mathbf{z}})\cos(\omega t))\cdot(\hat{\mathbf{z}}ds)\\ &= 3B_0\cos(\omega t)\int_S ds = 3B_0 ab\cos(\omega t).\end{aligned}$$

(b) Find a formula for the emf induced in the loop at time t. Simplify as much as possible. (6 points)

Here, the total emf is the transformer emf

$$\begin{aligned}V_{\mathrm{emf}} = V_{\mathrm{emf}}^{\mathrm{tr}} &= -N\frac{d\Phi}{dt} = -N\frac{d}{dt}(3B_0 ab\cos(\omega t)) = -3NB_0\,ab\frac{d}{dt}\cos(\omega t)\\ &= -3NB_0\,ab(-\omega\sin(\omega t)) = 3NB_0\,ab\,\omega\sin(\omega t).\end{aligned}$$

(c) Determine the current flowing through a $2\,k\Omega$ resistor incorporated in the loop when $N = 10$, $a = 5$ cm, $b = 2$ cm, $B_0 = 0.04$ T, $\omega = 4\times10^6$ rad/s, and $\omega t = \pi/6$, assuming that the resistance of the loop material is negligible. (8 points)

The angular frequency was initially incorrectly specified as $\omega = 4\,\mathrm{MHz}$. The correct units are 10^6 rad/s so that $\omega = 4\times10^6$ rad/s. If you thought I meant $f = 4\,\mathrm{MHz}$ and used $\omega = 8\pi\times10^6$ rad/s, I will accept a correct answer using this value.

In this case,

$$\begin{aligned}V_{\mathrm{emf}} &= 3(10)(0.04\,\mathrm{T})(5\times10^{-2}\,\mathrm{m})(2\times10^{-2}\,\mathrm{m})\left(4\times10^6\,\mathrm{rad\,s^{-1}}\right)\sin\left(\frac{\pi}{6}\right)\\ &= 0.24\times10^4\,\mathrm{Tm^2s^{-1}} = 2400\,\mathrm{V}.\end{aligned}$$

So,

$$I = \frac{V_{\mathrm{emf}}}{R} = \frac{2400\,\mathrm{V}}{2000\,\Omega} = 1.2\,\mathrm{A}.$$

2. A straight conducting wire of length $a = 20\,\text{cm}$ lying along the $\hat{\mathbf{x}}$ direction moves with velocity

$$\mathbf{u} = (2\hat{\mathbf{y}} + 3\hat{\mathbf{z}})\ \text{cm/s}$$

in a magnetic field

$$\mathbf{B} = (-0.1\hat{\mathbf{x}} + 0.2\hat{\mathbf{z}})\ \text{T}.$$

Find the emf between the two ends of the wire. (15 points)

In this case, the total emf V_{emf} is the motional emf

$$V_{\text{emf}} = V_{\text{emf}}^{\text{m}} = \int_\ell (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell}.$$

Here, $\mathbf{u} = (2\hat{\mathbf{y}} + 3\hat{\mathbf{z}})$ cm/s and

$$\mathbf{u} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0.02\,\text{m/s} & 0.03\,\text{m/s} \\ -0.1\,\text{T} & 0 & 0.2\,\text{T} \end{vmatrix} = (4\hat{\mathbf{x}} - 3\hat{\mathbf{y}} + 2\hat{\mathbf{z}}) \times 10^{-3}\,\text{Tm/s}$$

and $d\boldsymbol{\ell} = \hat{\mathbf{x}}\,d\ell$ so that

$$\begin{aligned}(\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} &= \big((4\hat{\mathbf{x}} - 3\hat{\mathbf{y}} + 2\hat{\mathbf{z}}) \times 10^{-3}\,\text{Tm/s}\big) \cdot (\hat{\mathbf{x}}\,d\ell) \\ &= \big(4 \times 10^{-3}\,\text{V/m}\big)\,d\ell.\end{aligned}$$

So,

$$V_{\text{emf}} = \big(4 \times 10^{-3}\,\text{V/m}\big) \int_\ell d\ell = \big(4 \times 10^{-3}\,\text{V/m}\big)\,(0.2\,\text{m}) = 0.8\,\text{mV}.$$

3. Suppose a total conduction current $I_c = 10\sin(\omega t)\,\mu\text{A}$ flows through a column of ammonia solution with cross-sectional area A in a glass tube. The solution has conductivity $\sigma \approx 10^{-3}$ S/m and relative permittivity $\epsilon_r \approx 10\pi$ and the current flowing through the glass is negligible compared to the current flowing through the solution. (10 points total)

   (a) Use the relationship between the conduction current, the current density, the cross-sectional area, and Ohm's law to determine the electric field intensity E at a cross-section of the solution as a function of the cross-sectional area. (5 points)

Here,

$$I_c = JA = \sigma EA$$

so that

$$E = \frac{I_c}{\sigma A} = \frac{(10^{-5}\,\mathrm{A})\sin(\omega t)}{(10^{-3}\,\mathrm{S/m})\,A} = \left(10^{-2}\,\mathrm{Am/S}\right)\frac{\sin(\omega t)}{A} = \frac{10^{-2}\sin(\omega t)}{A}\,\mathrm{V/m}.$$

(b) Determine the displacement current I_d through the cross-section when the angular frequency of I_c is $\omega = 2\times 10^8\,\mathrm{rad\,s^{-1}}$ and $\omega t = \pi/4$. (5 points)

The displacement current is

$$\begin{aligned}
I_d = J_d A = A\frac{\partial D}{\partial t} &= A\epsilon\frac{\partial E}{\partial t} = A\epsilon\frac{\partial}{\partial t}\left(\frac{10^{-2}\sin(\omega t)}{A}\,\mathrm{V/m}\right) \\
&= \epsilon_r\epsilon_0\left(10^{-2}\,\mathrm{V/m}\right)\omega\cos(\omega t) \\
&= (10\pi)\left(\frac{1}{36\pi}\times 10^{-9}\,\mathrm{F/m}\right)\left(10^{-2}\,\mathrm{V/m}\right)\left(2\times 10^8\,\mathrm{rad\,s^{-1}}\right)\cos(\pi/4) \\
&= \left(\frac{5}{9}\times 10^{-3}\,\mathrm{VF/(sm^2)}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{5\sqrt{2}}{18}\times 10^{-3}\,\mathrm{A} \approx 0.3928\,\mathrm{mA}.
\end{aligned}$$

4. An ammonia solution has a conductivity $\sigma \approx 10^{-3}$ S/m and $\epsilon_r \approx 10\pi$. (10 points total)

(a) Find the relaxation time constant for the solution. (5 points)

The relaxation time constant is

$$\tau_r = \frac{\epsilon}{\sigma} = \frac{\epsilon_r\epsilon_0}{\sigma} = \frac{(10\pi)\left(\frac{1}{36\pi}\times 10^{-9}\,\mathrm{F/m}\right)}{10^{-3}\,\mathrm{S/m}} = \frac{5}{18}\times 10^{-6}\,\mathrm{s} = 2.7778\times 10^{-7}\,\mathrm{s}.$$

(b) How long does it take for an excess charge density in the solution to decay to e^{-4} times its initial value? (5 points)

This happens when

$$\rho_{V0}e^{-t/\tau_r} = \rho_{V0}e^{-4},$$

which means

$$t = 4\tau_r = 1.1111\times 10^{-6}\,\mathrm{s}.$$

5. (25 points total) Suppose the electric field intensity of an EM wave in a nonconducting, source-free medium with $\epsilon_r = 9$ and $\mu_r = 1$ is

$$\mathbf{E}(z,t) = 6\cos\big(\big(4\pi \times 10^7\,\mathrm{rad/s}\big)\,t - kz + \pi/3\big)\hat{\mathbf{y}}\ \mathrm{V/m}.$$

(a) Determine the angular frequency ω, the wavenumber k, the wavelength λ, and the intrinsic impedance η. (10 points)

The angular frequency can be read from the electric field intensity and is

$$\omega = 4\pi \times 10^7\ \mathrm{rad/s}.$$

The wavenumber is

$$\begin{aligned} k &= \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_r\epsilon_r}\sqrt{\mu_0\epsilon_0} = \frac{\omega\sqrt{\mu_r\epsilon_r}}{c} \\ &= \frac{(4\pi \times 10^7\,\mathrm{rad/s})\,\sqrt{9}}{3 \times 10^8\,\mathrm{m/s}} = 0.4\pi\,\mathrm{rad/m} \approx 1.2566\,\mathrm{rad/m}, \end{aligned}$$

the wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.4\pi\,\mathrm{m}^{-1}} = 5\,\mathrm{m},$$

and the intrinsic impedance is

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r}{\epsilon_r}}\sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{1}{9}}\,\eta_0 = \frac{1}{3}(120\pi\,\Omega) = 40\pi\,\Omega \approx 125.66\,\Omega.$$

(b) Find $\tilde{\mathbf{E}}(z)$. (5 points)

Here,

$$\begin{aligned} \mathbf{E}(z,t) &= (6\,\mathrm{V/m})\cos\left(\omega t - kz + \frac{\pi}{3}\right)\hat{\mathbf{y}} = \mathrm{Re}\left\{(6\,\mathrm{V/m})\hat{\mathbf{y}}e^{j(\omega t - kz + \pi/3)}\right\} \\ &= \mathrm{Re}\left\{\left((6\,\mathrm{V/m})e^{j\pi/3}e^{-jkz}\hat{\mathbf{y}}\right)e^{j\omega t}\right\} \\ &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(z)e^{j\omega t}\right\}, \end{aligned}$$

where

$$\tilde{\mathbf{E}}(z) = (6\,\mathrm{V/m})e^{j\pi/3}e^{-jkz}\hat{\mathbf{y}}.$$

(c) Find $\tilde{\mathbf{H}}(z)$. (5 points)

In this case, the wave "travels" in the $+z$ direction and

$$\begin{aligned}\tilde{\mathbf{H}}(z) &= \frac{1}{\eta}\hat{k} \times \tilde{\mathbf{E}}(z) = \frac{1}{40\pi\,\Omega}(\hat{\mathbf{z}} \times \tilde{\mathbf{E}}(z)) = \frac{1}{40\pi\,\Omega}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ 0 & (6\,\mathrm{V/m})e^{j\pi/3}e^{-jkz} & 0 \end{vmatrix} \\ &= -\frac{6\,\mathrm{V/m}}{40\pi\,\Omega}\,e^{j\pi/3}e^{-jkz}\hat{\mathbf{x}} = -\left(\frac{3}{20\pi}\,\mathrm{A/m}\right)\,e^{j\pi/3}e^{-jkz}\hat{\mathbf{x}} \\ &\approx -(47.746\,\mathrm{mA/m})\,e^{j\pi/3}e^{-jkz}\hat{\mathbf{x}}.\end{aligned}$$

(d) Determine the average power density that the EM wave carries. (5 points)

Since the medium is lossless, the average power density is

$$S_{\mathrm{av}} = \frac{|\tilde{\mathbf{E}}|^2}{2\eta}\hat{\mathbf{z}}.$$

Here,

$$|\tilde{\mathbf{E}}|^2 = \left|(6\,\mathrm{V/m})e^{j\pi/3}e^{-jkz}\hat{\mathbf{y}}\right|^2 = |6\,\mathrm{V/m}|^2\left|e^{j(-kz+\pi/3)}\right|^2 = 36\,\mathrm{V^2/m^2}.$$

So,

$$S_{\mathrm{av}} = \frac{36\,\mathrm{V^2/m^2}}{80\pi\,\Omega}\hat{\mathbf{z}} = \frac{9\,\mathrm{VA/m^2}}{20\pi}\hat{\mathbf{z}} \approx 0.14324\hat{\mathbf{z}}\,\mathrm{W/m^2}.$$

6. Consider a $3/\pi$ GHz EM wave traveling in an ammonia solution, where $\sigma \approx 10^{-3}$ S/m, $\epsilon_r \approx 10\pi$, and $\mu_r = 1$. The wave travels downward into the solution, where the x-y plane is the surface, the $+z$ direction is down, and $z = 0^+$ is just below the surface. (20 points total)

(a) Find ϵ''/ϵ'. (10 points)

Here,

$$\begin{aligned}\omega &= 2\pi f = (2\pi\,\mathrm{rad})\left(3/\pi \times 10^9\right)\,\mathrm{s}^{-1} = 6 \times 10^9\,\mathrm{rad/s}, \\ \epsilon' &= \epsilon = \epsilon_r\epsilon_0 = (10\pi)\left(\frac{1}{36\pi} \times 10^{-9}\,\mathrm{F/m}\right) = \frac{5}{18} \times 10^{-9}\ \mathrm{F/m}, \\ \epsilon'' &= \frac{\sigma}{\omega} = \frac{10^{-3}\,\mathrm{S/m}}{(6 \times 10^9\,\mathrm{rad/s})} = \frac{1}{6} \times 10^{-12}\,\mathrm{Ss/m} = \frac{1}{6} \times 10^{-12}\,\mathrm{F/m}.\end{aligned}$$

So,

$$\frac{\epsilon''}{\epsilon'} = \frac{(1/6 \times 10^{-12}\,\text{F/m})}{(5/18 \times 10^{-9}\ \text{F/m})} = \frac{3}{5} \times 10^{-3} = 6 \times 10^{-4}.$$

(b) What type of medium is the ammonia solution at this frequency? Justify your answer. (5 points)

Since

$$\frac{\epsilon''}{\epsilon'} \ll 1,$$

the solution is a low-loss dielectric at this frequency.

(c) Find the skin depth for the ammonia solution at this frequency. (5 points)

For a low-loss dielectric,

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} = \left(5 \times 10^{-4}\,\text{S/m}\right)\sqrt{\frac{\mu_r}{\epsilon_r}}\sqrt{\frac{\mu_0}{\epsilon_0}} = \left(5 \times 10^{-4}\,\text{S/m}\right)\frac{1}{\sqrt{10\pi}}\eta_0$$

$$= \left(5 \times 10^{-4}\,\text{S/m}\right)\frac{120\pi\,\Omega}{\sqrt{10\pi}} = (6 \times 10^{-3})\sqrt{10\pi}\,\text{Np/m}.$$

So, the skin depth is

$$\delta_s = \frac{1}{\alpha} \approx \frac{1}{(6 \times 10^{-3})\sqrt{10\pi}}\,\text{m} \approx 29.735\,\text{m}.$$